

Solid Connections

DESIGN REVIEW 4

41070 Embedded Mechatronics Studio

Team Introduction [Lab 2 Group 4]



Olivia Alvarez

Team/Project Lead

Filtering Subcircuit
Calibration Subroutine



Yuto Boittiaux

Mechanical Lead

Accelerometer Subcircuit
Self-Test Subroutine
Enclosure Codesigner



John Chen

Integration Lead

I/O & Battery Monitoring
Subcircuit
Walking-Pace Subroutine



Guy Smith

PCB Lead

Power Delivery Subcircuit
Step Tracker Subroutine
Enclosure Codesigner

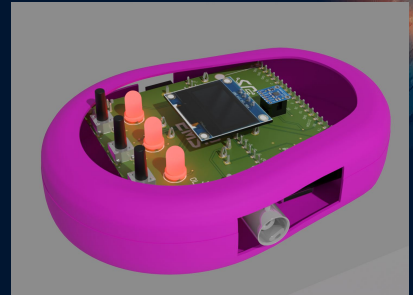
Step Tracker: System and Application Overview

Purpose

- To detect whether the user is *stationary*, *walking* or *running*
- Track the *number of steps* the user has taken
- Be an ergonomic and easy to use step tracker

System Parts

- ESP32 integrated with a custom PCB
- Accelerometer for movement tracking
- Mechanical enclosure imitating a tamagotchi for interactive user experience
- User interface to indicate step count and allow control of software modes



Project expectations

Contract

1. Attendance & Participation
2. Communication
3. Collaboration
4. Project Timeline
5. Task Tracking
6. Financial Contributions

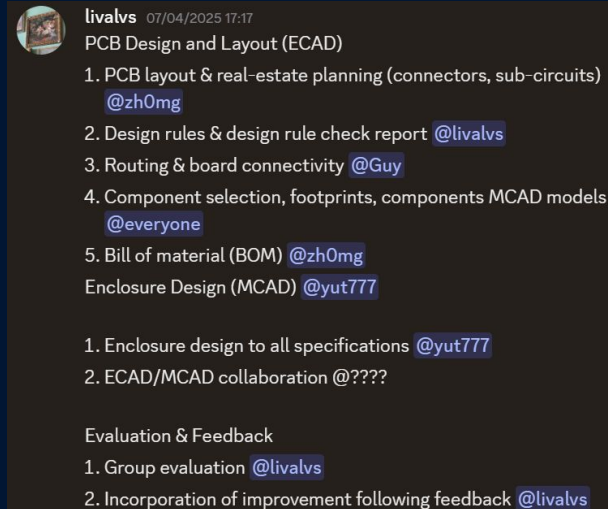
Communication and Collaboration

Primary Communication: Discord

Collaboration: Google Drive

Response Expectations: 24 hours

Task Tracking: Gantt Chart



A screenshot of a Discord message from user **livalvs** (07/04/2025 17:17) titled "PCB Design and Layout (ECAD)". The message contains a numbered list of tasks and their assignees, followed by a section for "Evaluation & Feedback".

livalvs 07/04/2025 17:17
PCB Design and Layout (ECAD)

1. PCB layout & real-estate planning (connectors, sub-circuits) @zh0mg
2. Design rules & design rule check report @livalvs
3. Routing & board connectivity @Guy
4. Component selection, footprints, components MCAD models @everyone
5. Bill of material (BOM) @zh0mg

Enclosure Design (MCAD) @yut777

1. Enclosure design to all specifications @yut777
2. ECAD/MCAD collaboration @????

Evaluation & Feedback

1. Group evaluation @livalvs
2. Incorporation of improvement following feedback @livalvs

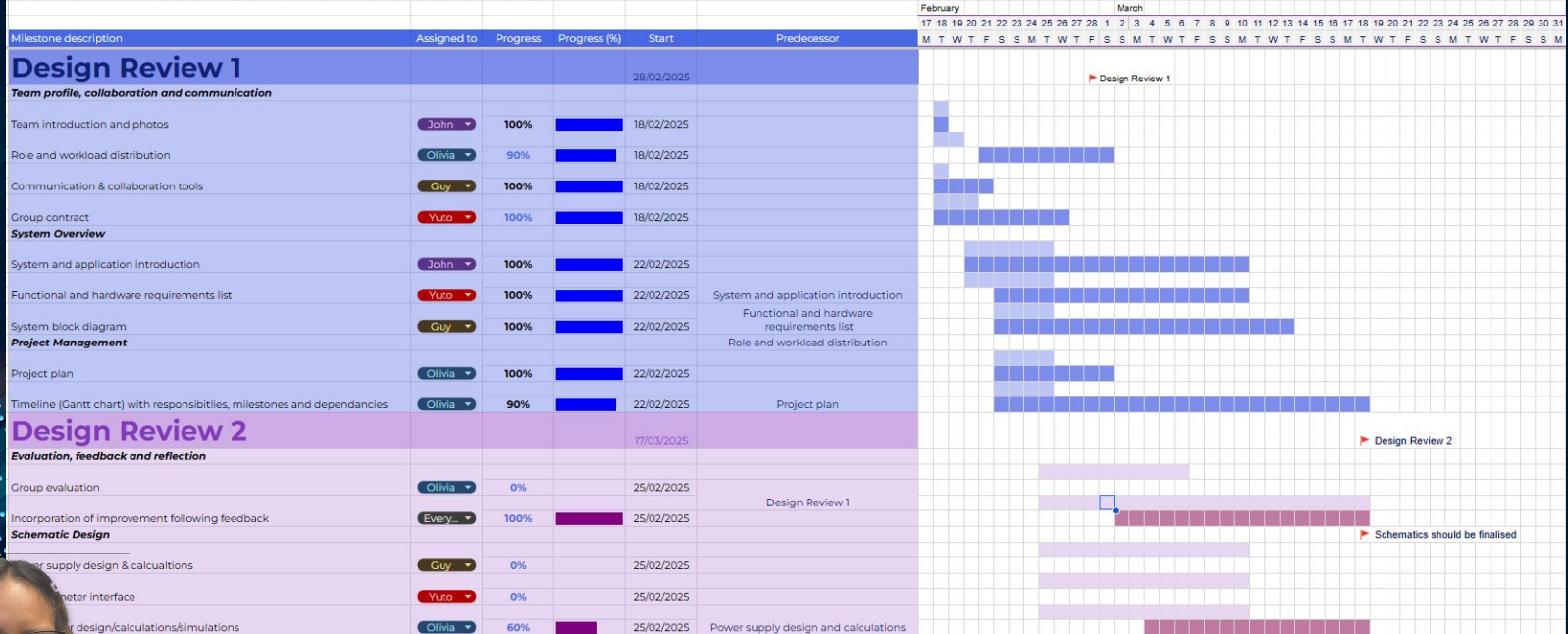


Project Plan / Gantt Chart

41070 Embedded Mechatronics Studio

Solid Connections

Group members: Olivia Alvarez, Yuto Boittiaux, John Chen, Guy Smith





System Overview and Specifications

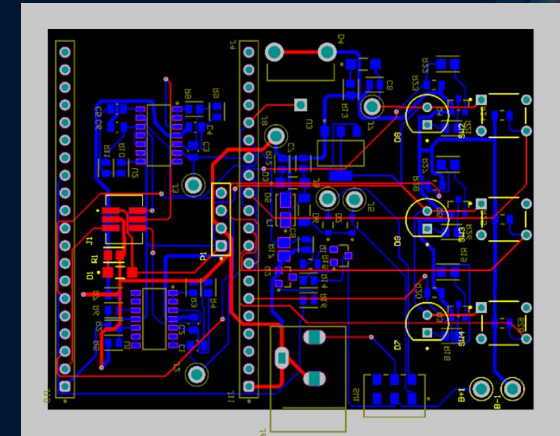
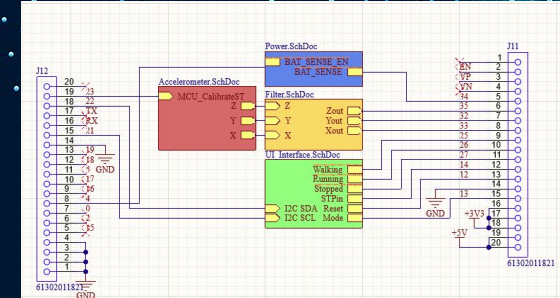
Functional and Hardware Requirements

Subsystem	Details
Power	2S LiPo (7.2 V) + external lab PSU via DC barrel jack. Charging circuit + 3.3 V LDO regulator. Voltage divider to ESP32 ADC. System switch and power indicator LEDs.
Sensor	ADXL335 with analog output (3.3 V), ST pin toggled via GPIO. Analog low-pass filtering on X, Y, Z axes.
I/O	Minitex127 headers, test points for power/filtering signals, BNC connector for Z-axis debug output.
User Interface	3x push buttons: Mode, Reset, Calibrate. 3x pace LEDs. OLED (128×64, I2C).
Microcontroller	ESP32 dev board handles acquisition, pace detection, calibration, display, and control.
Mechanical Enclosure	Tamagotchi-style housing with cutouts for I/O, wearable attachment, and mounting interface.



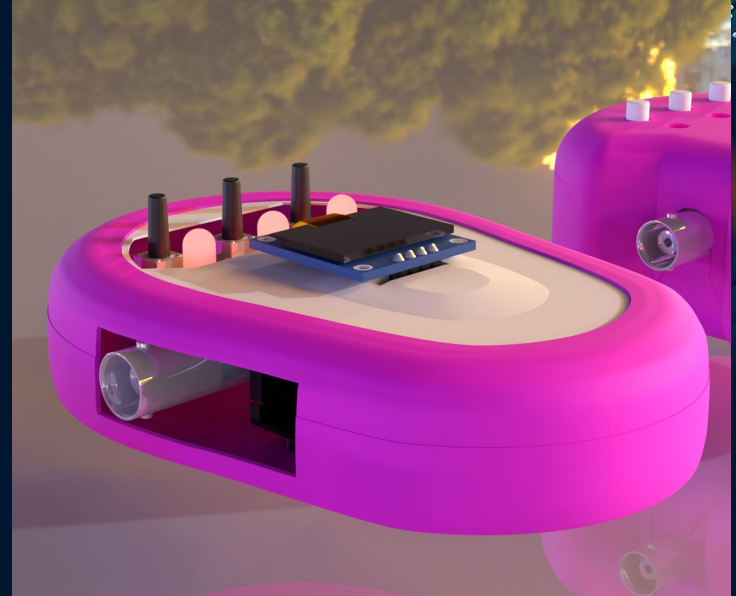
Project Evolution

Aspect	Design 1	Design 2	Design 3
Microcontroller	Arduino Uno/Nano	ESP32	ESP32
Power Supply	9V Battery	2 Lithium-Ion cells in series	2 Lithium-Ion cells in series
PCB Layers	Single-layer	Dual-layer	Dual-layer
Signal Conditioning	Basic second-order filter	Sallen-key second order filter with absence of buffer amplifier	Sallen-key second-order filter with buffer amplifier
User Interface	Not considered	Inclusion of Buttons and OLED	Full Implementation of buttons, OLED and LEDs
Connection Accuracy	Convolved connections and no microcontroller integration	Incorrect ADXL assignments and pin headers	Correct ADXL connection and ESP32 pin alignment
Mechanical Enclosure	Simple box casing	Tamagotchi-style concept ideated	Tamagotchi-style refined and made more ergonomic



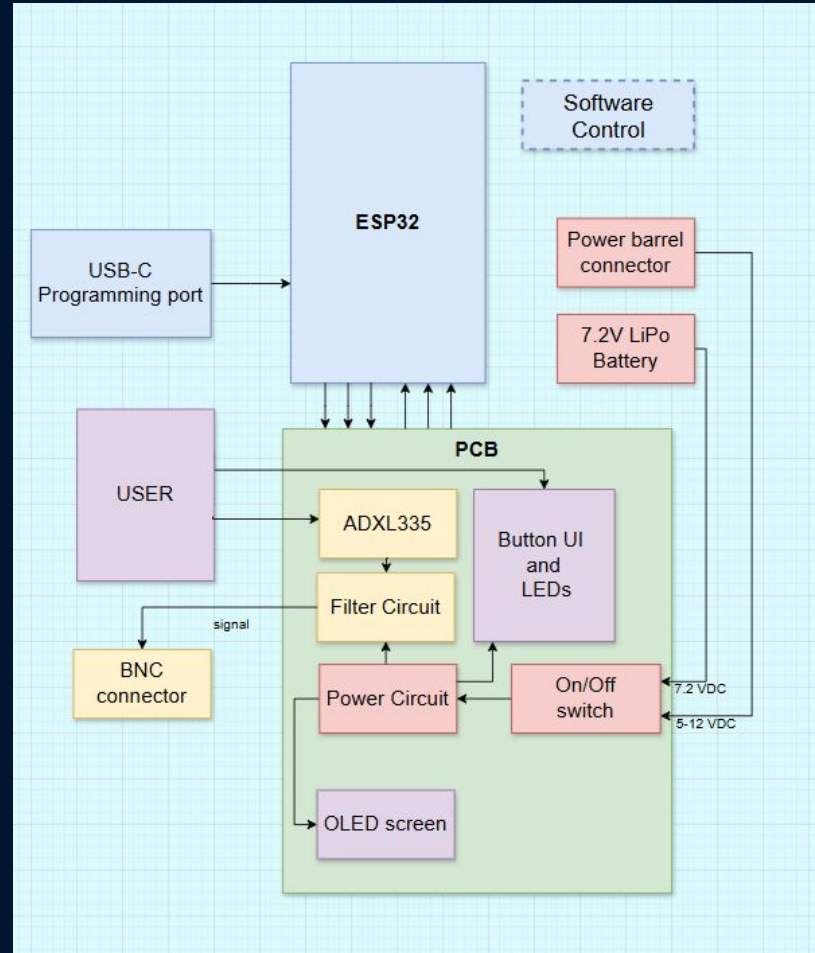
Finalised Concept

- ESP32 Microcontroller
- 7.6 LiPo Battery
- Sallen-Key second-order low-pass filter
- Tamagotchi style ergonomic mechanical enclosure
- OLED screen compatible with mechanical casing
- 3 buttons and 3 LEDs for better user experience



Block Diagram

Key	Descriptor
	PCB
	ESP32
	Software
	User experience
	Power





PCB Innovation and Creativity

Innovative PCB features

Double sided

- Thru-hole on top, smd on bottom

Attention to detail

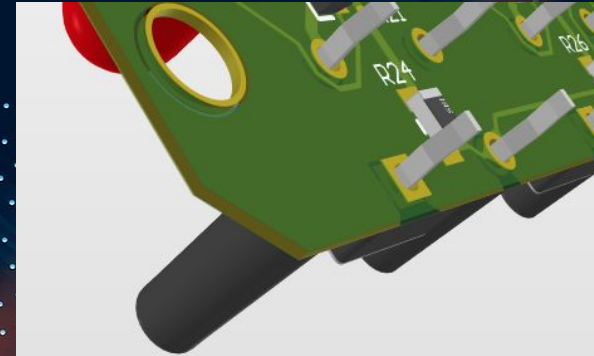
- All components placed using coordinates, not snaps.

Beveled edges

- Safety in mind, cannot be used as a box cutter

Low voltage protection

- Turns off instead of draining batteries too low



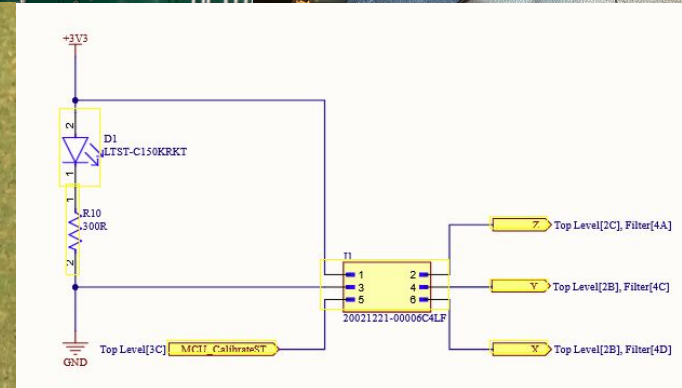
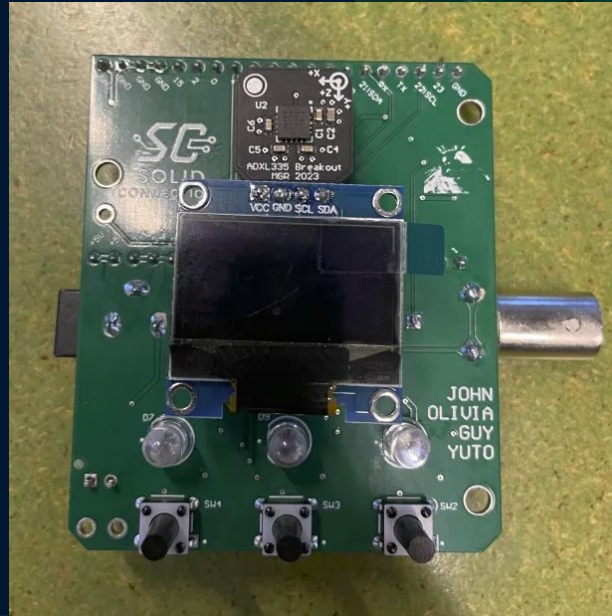
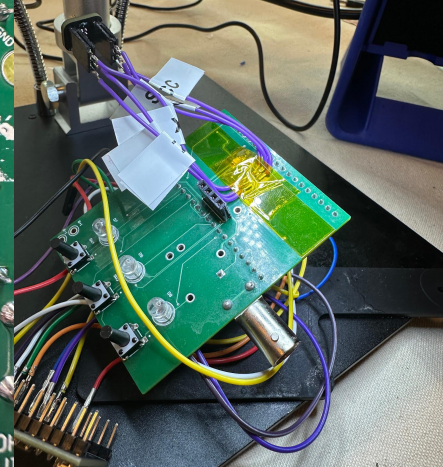
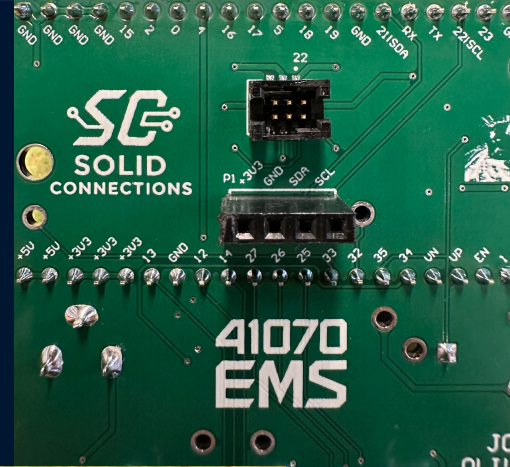
Creative Design: Accelerometer

The ADXL is mounted on the top to save space.

- Filter is on other side, under ESP

Notable changes :

- Fixed incorrect pin mountings
- Changed port to direct voltage



User Experience

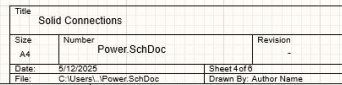
Big Tamagotchi
(Roughly 2.5x scale)



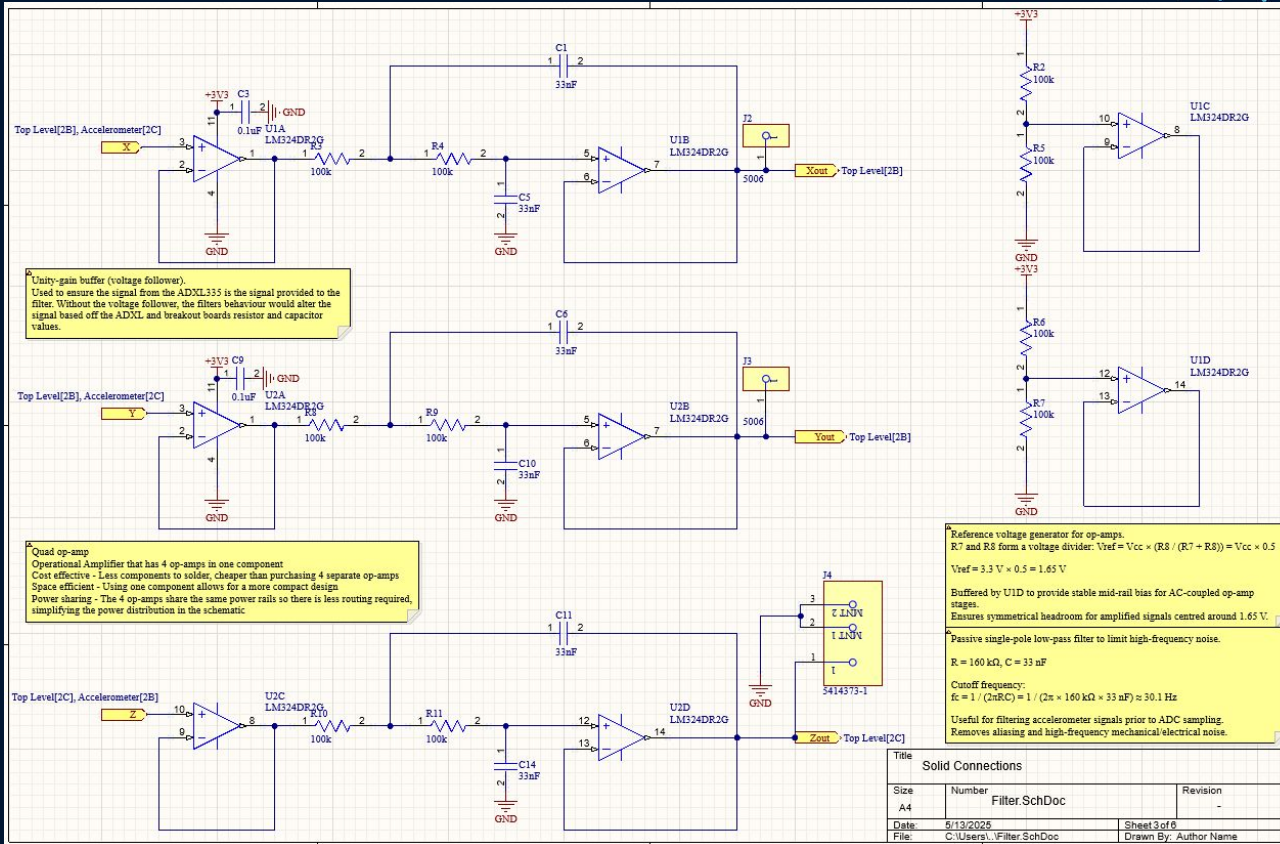
Buttons on bottom
Screen on top
Ergonomic
Fits in Pocket



-
- Barrel jack and alternate input (ACTIO) to power the system.** Diodes D1,D3 provide reverse polarity protection.
- Schottky diodes for reverse current protection.** Voltage drop: ~0.3~0.4 V. Prevent backfed from sources.
- LED current limiters.**
 $I_{LED} \approx I_{drop}(R_8 - R_9)$
 Assuming $\gamma = 1$, LED drop:
 $R_8 \Rightarrow \approx 0.4\text{ mA}/(0.4\text{mA} \cdot (\text{indicator}))$
 $R_9 \Rightarrow \approx 0.3\text{ mA}/(0.4\text{mA} \cdot (\text{brightness}))$
- Power switch—toggles power from battery to system input rail.**
- P-Channel MOSFET acts as a high-side switch for the voltage divider.** Turns on when gate is pulled low by Q2.
- Voltage divider:**
 $V_{AD} = V_{BAT} \cdot R_1 / (R_1 + R_2) \approx 5.6\text{V}$
 $k_1(V_{BAT} - 0.36)/V_{AD} = V_{BAT} \approx 108/5.65 \approx 19.1$
 At 8.4V battery → ADC reads $n \approx 3.02V$ (safe for ESP32)
- Base resistors:**
 $V_{GPIO} = 3V/V_{BE} \approx 0.7V \Rightarrow I_B = 3 - 0.7/1k \approx 2.3\text{ mA}$
 $I_V = I_B \times 10 \approx 23\text{ mA}$
 Safe for ESP32 GPIO and sufficient to saturate Q2.
- Regulator to power ESP32 and peripherals.** Input = VBAT (post-protection). Output = 3.3V.
- Input/output decoupling caps.** Stabilize voltage during transients.
- TVS diode for voltage spike protection.** Lets current flow to ground during spikes preventing damage to LDO and indicator LEDs.
- BAT_SENSE** — Top Level[C]
- Filtering capacitor for BAT_SENSE.** Reduces ADC noise.
 Formin low-pass filter with $R_1 = R_2$:
 $f_c = 1/(2\pi RC) \approx 284\text{ Hz}$; $f_{Hfc} = 2\pi C/1024\text{ Hz}$
- | Title
Solid Connections | | | |
|----------------------------|-----------------------|--------|-----------------------|
| Size | A4 | Number | Revision |
| Date: | 5/14/2005 | | Sheet 4 of 8 |
| File: | C:\Users\Power\SchDoc | | Drawn By: Author Name |



Analog Filter - Schematic



Key Notes:

Capacitor: 33nF ceramic

Resistor: 160kΩ

Op-Amp: 4-Circuit

Cut-off Frequency:
30Hz

Test Points on the X
and Y outputs



Analog Filter - Mathematics

$$f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

**Note that this is the formula for the resonant (natural) frequency*

$$R_1 = R_2 \Rightarrow R$$

$$C_1 = C_2 \Rightarrow C$$

$$\therefore f_c = \frac{1}{2\pi RC}$$

$$R = \frac{1}{2\pi f_c C}$$

With a capacitor value of 33nF

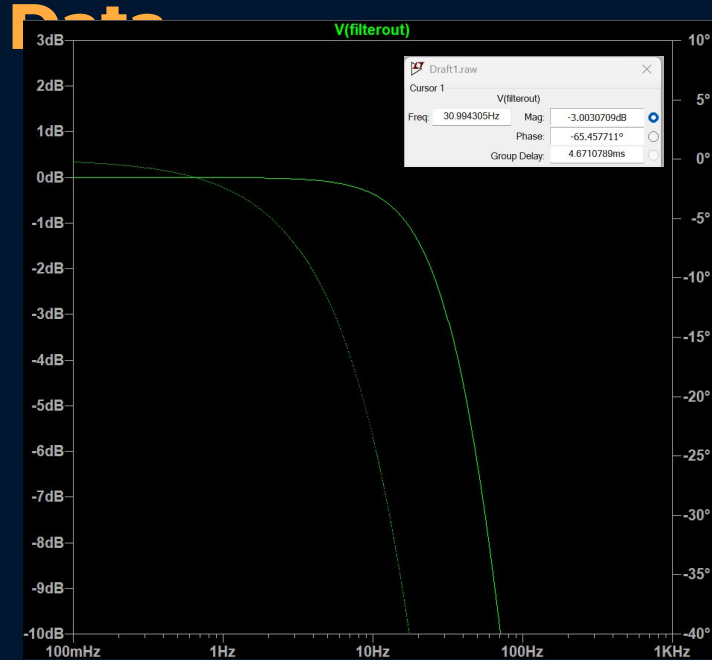
$$R = \frac{1}{2\pi(30)(33 \times 10^{-9})}$$

$$R = 160\,762\Omega$$

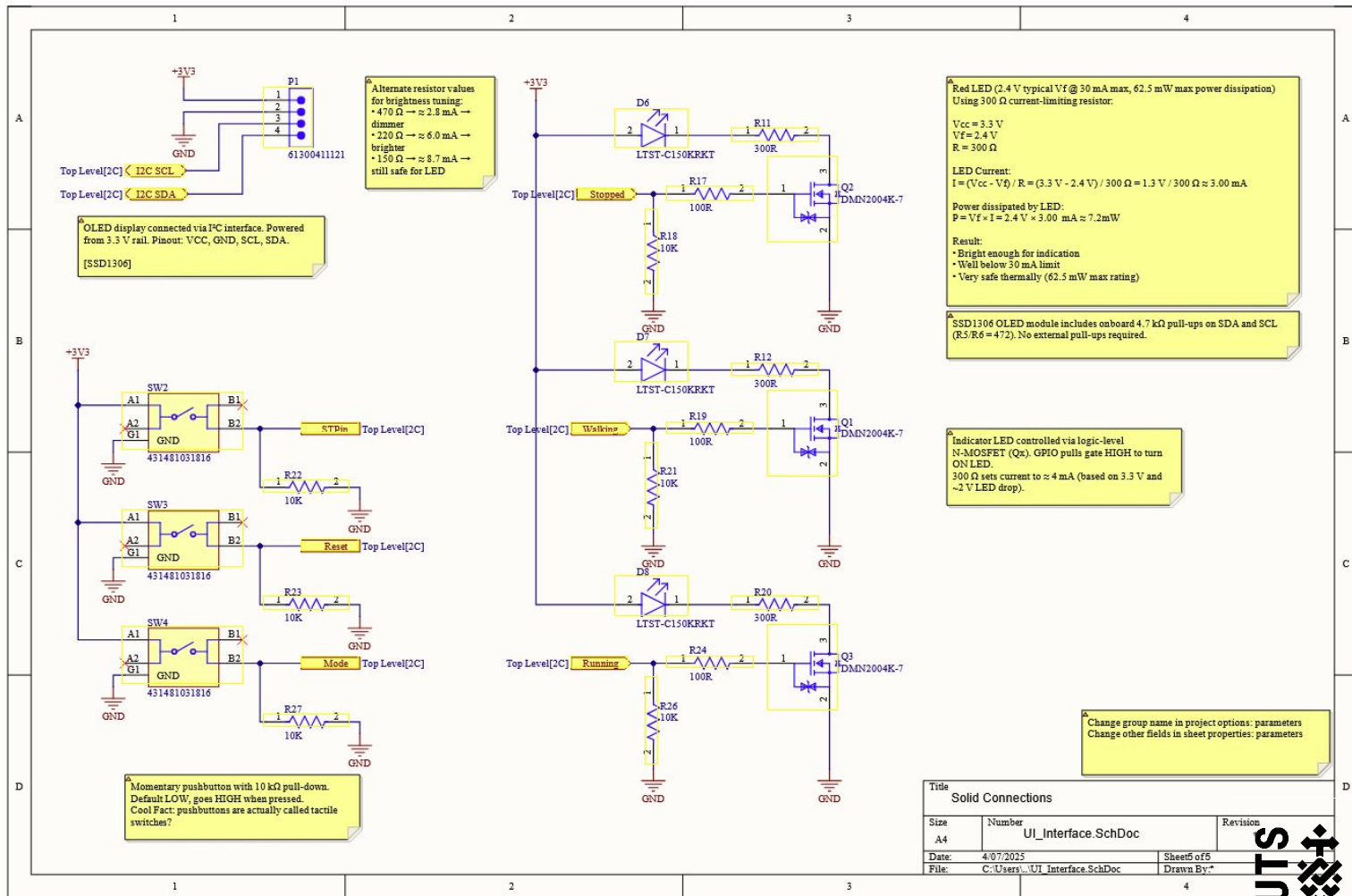
$$R \approx 160k\Omega$$



Analog Filter - Simulation vs Tested



Added Comments



Signal Mapping Table

Net Name	GPIO	Direction	Function / Notes
MCU_CalibrateST	23	Input	User button to trigger calibration/self-test routine
I2C_SDA	22	I/O	I2C data line for OLED display
I2C_SCL	21	I/O	I2C clock line for OLED display
BAT_SENSE_EN	4	Output	Enable voltage divider to read battery level via BAT_SENSE
BAT_SENSE	34	Analog In	ADC input to sense battery level through voltage divider
Zout	35	Analog In	Filtered Z-axis output from ADXL335 (also sent to BNC connector)
Yout	32	Analog In	Filtered Y-axis output from ADXL335
Xout	33	Analog In	Filtered X-axis output from ADXL335
Walking	25	Output	LED indicator for walking pace
Running	26	Output	LED indicator for running pace
Stopped	27	Output	LED indicator for stationary state
STPin	14	Output	ADXL335 self-test control pin
Reset	12	Input	User button to reset step counter
Mode	13	Input	User button to switch between operational modes (e.g., pace, display, etc.)



BOM

Line #	Name	Description	Designator	Revision ID	Revision State	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
27	20021221-00006C4	Minitek127*, Wire	J1		Not managed		1	Amphenol ICC / FC	20021221-00006C4LF	Unknown	Digikey	609-3699-1-ND	1.11	5.57
	PJ-102AH	Power Barrel Con.	J5		Not managed		1	CUI Devices	PJ102AH	Unknown	Digikey	CP-102AH-ND	1.26	6.32
11	DMN2004K-7	MOSFET N-CH 20V	Q3, Q4, Q5	CMP-12801-001	New From Design	Up to date	3	Diodes	DMN2004K-7	Volume Production	Digikey	DMN2004KDICT-ND	0.35604	5.34
	C0805C103J5RAC72	CAP CER SMD 0805	C9	CMP-2007-0111	New From Design	Up to date	1	KEMET	C0805C103J5RAC7210	Volume Production	Digikey	399-17616-1-ND	0.07487	0.74869
9	C0805C104J5RACTU	CAP CER 0.1UF 50V	C8	CMP-03020-001	New From Design	Up to date	1	KEMET	C0805C104J5RACTU	Volume Production	Digikey	399-C0805C104J5RACTU	0.24956	1.25
17	5006	PC TEST POINT CON	B-1, B+1, J2, J3, J4,	CMP-19636-001	New From Design	Up to date	8	Keystone Electroni	5006	Volume Production	Digikey	36-5006-ND	0.43657	17.46
2	08055C104JAZ2A		C7	CMP-2007-0431	Released	Up to date	1	Kyocera AVX	08055C104JAZ2A	Volume Production	Digikey	478-KGF21BR71H104JTCT-	0.58231	2.91
19	CL-SB-22A-11T	SWITCH SLIDE DPD	SW1	CMP-71745-001	New From Design	Up to date	1	Nidec Copal	CL-SB-22A-11T	Volume Production	Digikey	563-1340-1-ND	2.35	11.73
8	LM324DR2G	Single Supply Quad	U1, U3	CMP-1310-001	New From Design	Up to date	2	ON Semiconductor	LM324DR2G	Volume Production	Digikey	LM324DR2GOSCT-ND	0.66883	6.69
22	MMBT3904LT1G	General Purpose T	Q2	CMP-1048-0061	Released	Up to date	1	ON Semiconductor	MMBT3904LT1G	Volume Production	Digikey	MMBT3904LT1GOSCT-ND	0.19965	0.99825
21	ERJ-6GEYJ563V		R17	CMP-2001-0041	Released	Up to date	1	Panasonic	ERJ-6GEYJ563V	Volume Production	Digikey	P56KACT-ND	0.06988	0.69878
	RMCF0805FG100K	100kΩ ±1% 0.125W	R14	CMP-26527-001	New From Design	Up to date	1	Stackpole Electron	RMCF0805FG100K	Volume Production	Digikey	RMCF0805FG100KCT-ND	0.04159	0.41594
	RMCF0805FT1K00	1kΩ ±1% 0.125W	R16	CMP-26527-001	New From Design	Up to date	1	Stackpole Electron	RMCF0805FT1K00	Volume Production	Digikey	RMCF0805FT1K00CT-ND	0.04159	0.41594
	RMCF0805FT10K0	RES 10K OHM 1% 1/10	R9, R11, R15, R20,	CMP-26527-001	New From Design	Up to date	7	Stackpole Electron	RMCF0805FT10K0	Volume Production	Digikey	RMCF0805FT10K0CT-ND	0.04159	1.46
	RMCF0805FT100R	100Ω ±1% 0.125W	R19, R22, R27	CMP-26527-001	New From Design	Up to date	3	Stackpole Electron	RMCF0805FT100R	Volume Production	Digikey	RMCF0805FT100RCT-ND	0.04159	0.62391
	RMCF0805JT160K	160kΩ ±5% 0.125W	R1, R2, R3, R4, R5,	CMP-26527-001	New From Design	Up to date	10	Stackpole Electron	RMCF0805JT160K	Volume Production	Digikey	RMCF0805JT160KCT-ND	0.02096	1.05
13	RMCF0805JT3K00	3kΩ ±5% 0.125W	R12	CMP-26527-001	New From Design	Up to date	1	Stackpole Electron	RMCF0805JT3K00	Volume Production	Digikey	RMCF0805JT3K00CT-ND	0.0366	0.36603
14	RMCF0805JT300R	300Ω ±5% 0.125W	R1, R13, R18, R21,	CMP-26527-001	New From Design	Up to date	5	Stackpole Electron	RMCF0805JT300R	Volume Production	Digikey	738-RMCF0805JT300RCT-N	0.0366	0.91506
7	LD1117S33CTR	IC REG LINEAR 3.3V	U2	CMP-0244-0031	New From Design	Up to date	1	STMicroelectronics	LD1117S33CTR	Volume Production	Digikey	497-1241-1-ND	0.5324	2.66
10	CUS08F30,H3F	CUS08F30,H3F Rect	D2, D5	CMP-00375-531	New From Design	Up to date	2	Toshiba	CUS08F30,H3F	Volume Production	Digikey	CUS08F30H3FCT-ND	0.25622	2.56
24	SSM3J328R,LF	TOSHIBA Field-Effe	Q1	CMP-2100-0361	Released	Up to date	1	Toshiba	SSM3J328R,LF	Volume Production	Digikey	SSM3J328RLFCT-ND	0.68214	3.41
	LTST-C150KRKT	LED RED CLEAR 120	D1, D3, D4, D6, D7,	CMP-2000-0661	New From Design	Up to date	6	Vishay Lite-On	LTST-C150KRKT	Volume Production	Digikey	160-1405-6-ND	0.20797	6.24
3	61300411121		P1	CMP-1502-0101	Released	Up to date	1	Würth Electronics	61300411121	Volume Production	Digikey	732-5317-ND	0.28284	1.41
16	61302011821		J9, J10	CMP-1502-0131	Released	Up to date	2	Würth Electronics	61302011821	Unknown	Digikey	732-61302011821-ND	1.38	13.81
18	431481031816	Tactile Switch, 0.05	SW2, SW3, SW4	CMP-1457-0001	Released	Up to date	3	Würth Electronics	431481031816	Volume Production	Digikey	732-7038-1-ND	0.56568	8.49
	CC0805KRX7R9BB3	CAP CER 0.033UF 50V	C1, C2, C3, C4, C5,	CMP-03422-001	New From Design	Up to date	6	Yageo	CC0805KRX7R9BB333	Volume Production	Digikey	311-1374-1-ND	0.09816	2.94

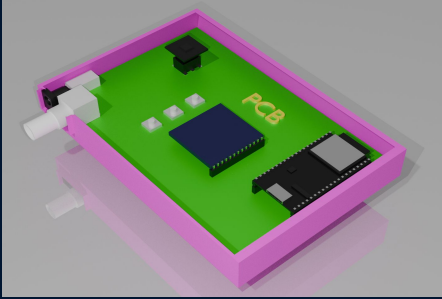


Enclosure Innovation and Creativity



Enclosure Design

Prototype 1



Prototype 2



Design Requirements:

Sturdy enclosure to house the PCB stack with appropriate use of fasteners.

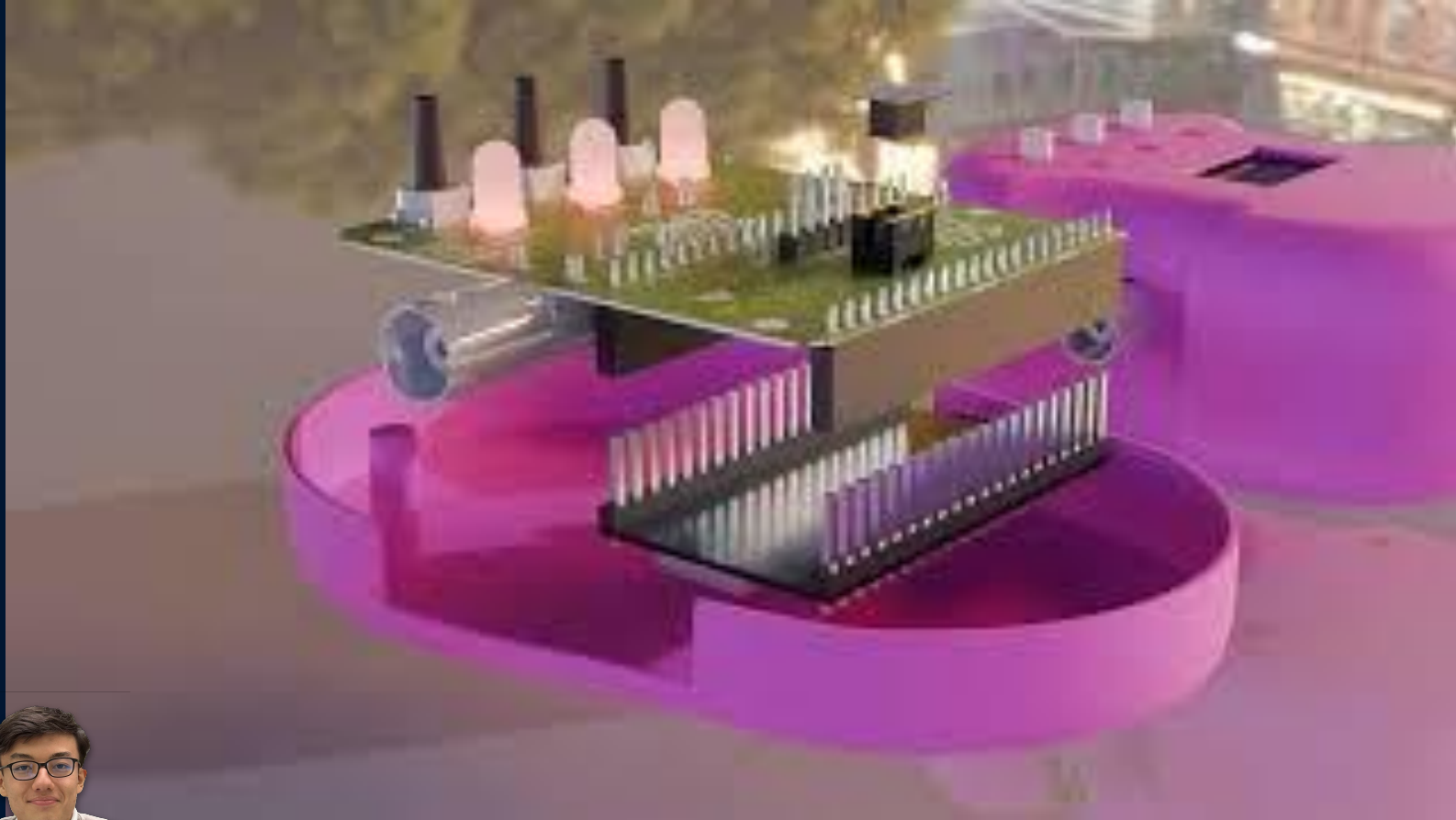
Appropriately sized cut-outs for peripherals (LEDs, buttons, switches, LCD) and cable connections.

Provide means to attach the enclosure to your body to demonstrate the step tracker.

Provide a mounting interface for the experimental demonstration and test setup. Drawing provided below.

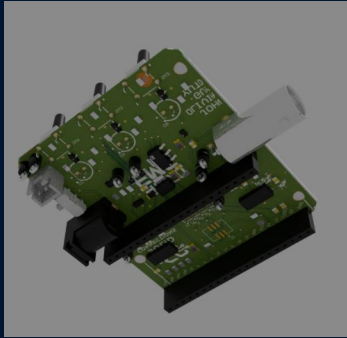


Enclosure Design Evolution



Enclosure Design

Prototype 3



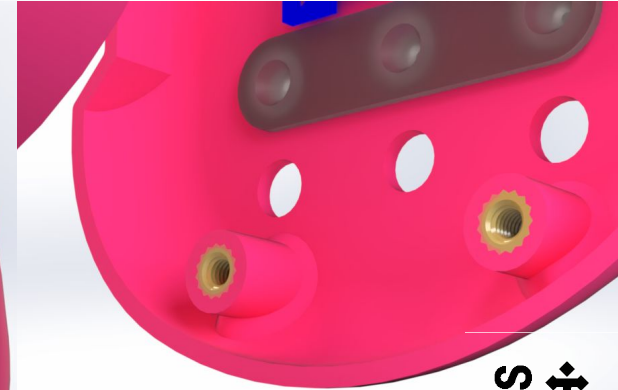
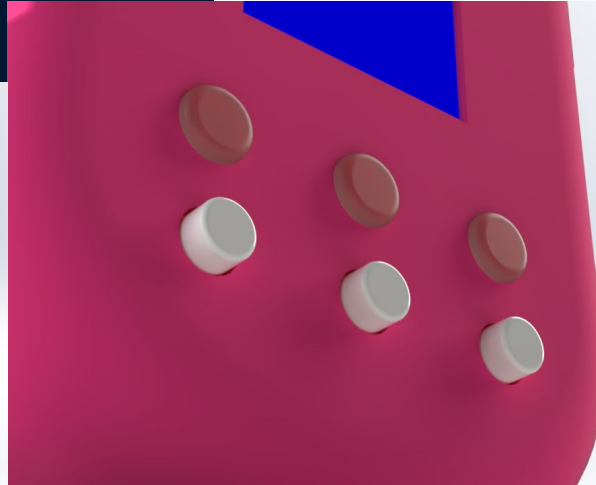
Adapted enclosure to the new board design
as our PCB was now double sided



Enclosure Design

Prototype 4

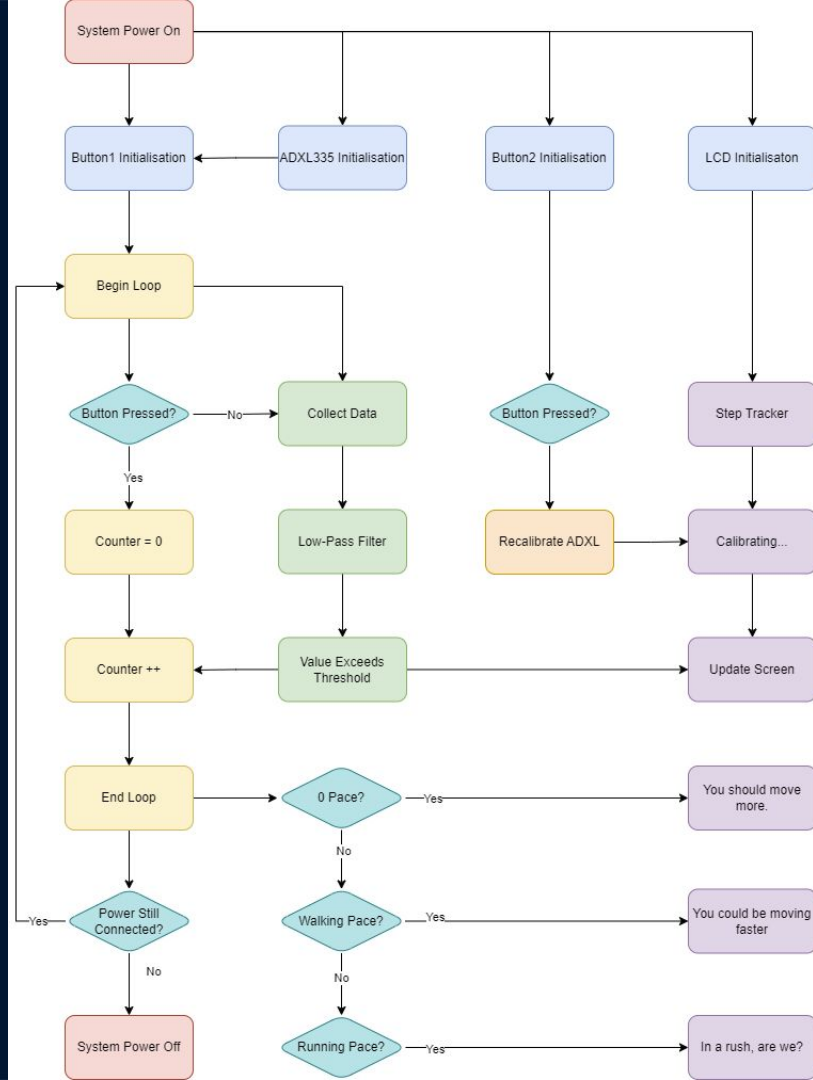
1. Added M3 Inserts for mounting
2. Added button caps for ergonomics
3. Added a LED diffuser for aesthetics
4. Refined geometry for design intent
5. Incorporated embossed logo for branding
6. Considered battery mounting
7. Designed for easy assembly
8. Added USB port cover





Embedded Software Innovation and Creativity

Flowchart of Step Tracker program



Self-test routine

Once the ST button (left button) is pressed, the following outputs should be observed:

- **X-axis:** Decreases by $\sim 1.08g$ ($\sim -325mV$)
- **Y-axis:** Increases by $\sim 1.08g$ ($\sim +325mV$)
- **Z-axis:** Increases by $\sim 1.83g$ ($\sim +550mV$)

Knowing this, we'll be able to display if the testing works and therefore check if the Accelerometer itself works.

```
void setup() {  
  Serial.begin(9600);  
  pinMode(ST_PIN, OUTPUT);  
  digitalWrite(ST_PIN, LOW);  
  delay(1000); // Allow sensor to stabilize  
}  
  
void loop() {  
  // Measure baseline (self-test OFF)  
  digitalWrite(ST_PIN, LOW);  
  delay(100);  
  float xBaseline = readAcceleration(X_PIN, V_ZERO_G_X);  
  float yBaseline = readAcceleration(Y_PIN, V_ZERO_G_Y);  
  float zBaseline = readAcceleration(Z_PIN, V_ZERO_G_Z);  
  
  // Activate self-test  
  digitalWrite(ST_PIN, HIGH);  
  delay(100);  
  float xTest = readAcceleration(X_PIN, V_ZERO_G_X);  
  float yTest = readAcceleration(Y_PIN, V_ZERO_G_Y);  
  float zTest = readAcceleration(Z_PIN, V_ZERO_G_Z);  
  
  // Calculate differences in g  
  float xDiffG = xTest - xBaseline;  
  float yDiffG = yTest - yBaseline;  
  float zDiffG = zTest - zBaseline;  
  float xDiffmV = xDiffG * SENSITIVITY;  
  float yDiffmV = yDiffG * SENSITIVITY;  
  float zDiffmV = zDiffG * SENSITIVITY;  
  
  // Check self-test against datasheet ranges  
  if (checkSelfTest(xDiffG, yDiffG, zDiffG, xDiffmV, yDiffmV, zDiffmV)) {  
    Serial.println("Self-Test: PASS");  
  } else {  
    Serial.println("Self-Test: FAIL");  
  }  
  
  delay(5000); // Wait before repeating  
}
```



Calibration routine

- Starts when user presses the calibration button
- Reads sensor data for each position
- Converts voltage to G-force values
- Calculates average offset for each axis
- Shows success message on OLED when done

```
void runCalibration() {
    display.clearDisplay();
    display.setCursor(0, 0);
    display.print("Calibrating...");
    display.display();
    Serial.println("Starting Calibration...");

    AxisData calibrationSample = {0, 0, 0};
    const int numSamples = 100;
    for (int i = 0; i < numSamples; i++) {
        AxisData currentReading = readAllAxes();
        calibrationSample.x += currentReading.x;
        calibrationSample.y += currentReading.y;
        calibrationSample.z += currentReading.z;
        delay(10);
    }

    float x_offset = calibrationSample.x / numSamples;
    float y_offset = calibrationSample.y / numSamples;
    float z_offset = calibrationSample.z / numSamples;

    Serial.println("Calibration Complete");
    Serial.print("X Offset: "); Serial.println(x_offset, 3);
    Serial.print("Y Offset: "); Serial.println(y_offset, 3);
    Serial.print("Z Offset: "); Serial.println(z_offset, 3);

    display.clearDisplay();
    display.setCursor(0, 0);
    display.print("Calibration OK");
    display.setCursor(0, 20);
    display.print("X Offset: "); display.print(x_offset, 3);
    display.setCursor(0, 30);
    display.print("Y Offset: "); display.print(y_offset, 3);
    display.setCursor(0, 40);
    display.print("Z Offset: "); display.print(z_offset, 3);
    display.display();
    delay(2000);
}

AxisData readAllAxes() {
    return {
        voltageToG(analogRead(X_PIN)),
        voltageToG(analogRead(Y_PIN)),
        voltageToG(analogRead(Z_PIN))
    };
}

float voltageToG(int raw) {
    float voltage = (raw / (float)ADC_RESOLUTION) * ADC_REF_VOLTAGE;
    return (voltage - 2.5) / 0.33;
}
```



Step tracker routine

Initialise System – Set up peripherals and check connections.

Read Accelerometer Data – Capture real-time acceleration values.

Filter Data – Remove noise using a band-pass filter.

Detect Steps – Identify movement patterns from accelerometer data.

Count Steps – Increment step counter for each valid step.

Determine Walking Pace – Classify motion as stationary, walking, or running.

Update Display – Show step count and pace on the LCD.

Check User Input – Reset counter or start calibration if a button is pressed.

```
60 void loop() {
61   // Battery status
62   int batteryADC = analogRead(BATTERY_ADC);
63   float batteryVoltage = (batteryADC / (float)ADC_RESOLUTION) * ADC_REF_VOLTAGE * BATTERY_DIVIDER_RATIO;
64   int batteryPercent = map(batteryVoltage * 100, 330, 420, 0, 100);
65   batteryPercent = constrain(batteryPercent, 0, 100);
66
67
68   // Step tracking
69   runStepTracker();
70
71   // Show step count + pace + battery
72   displayStepInfo(stepCount, getCurrentPace(), batteryPercent);
73
74   // Run calibration/self-test
75   if (digitalRead(BTN_CALIBRATE) == LOW) {
76     runSelfTest();
77     runCalibration();
78     delay(1000);
79   }
80
81   delay(100);
82 }
83
84 void runStepTracker() {
85   AxisData data = readAllAxes();
86   float magnitude = sqrt(data.x * data.x + data.y * data.y + data.z * data.z);
87   unsigned long currentTime = millis();
88
89   if (magnitude > stepThreshold && (currentTime - lastStepTime > debounceTime)) {
90     lastStepTime = currentTime;
91     stepCount++;
92   }
93
94   if (digitalRead(BTN_RESET) == LOW) {
95     displayMessage("Resetting...");
96     while (digitalRead(BTN_RESET) == LOW); // wait for release
97     stepCount = 0;
98     displayMessage("Steps Reset");
99     delay(1000);
100   }
101 }
102
```



Walking-pace identification routine

Step Frequency Analysis

- Calculate the time between detected steps.
- Identify patterns that indicate walking or running (e.g., frequency of step bursts).

Classify Pace

- **Stationary:** No steps detected or very slow frequency.
- **Walking:** Step frequency falls within a defined range (e.g., 1-2 Hz).
- **Running:** Higher step frequency (e.g., > 2 Hz).

Update Display

- Show the corresponding activity on the LCD (Stationary, Walking, Running).

```
110 // Determine pace based on step frequency
111 const char* pace;
112 if (stepFrequency < 0.5) {
113     pace = "Stationary";
114     digitalWrite(LED_STOPPED, HIGH);
115     digitalWrite(LED_WALKING, LOW);
116     digitalWrite(LED_RUNNING, LOW);
117 } else if (stepFrequency <= 2.0) {
118     pace = "Walking";
119     digitalWrite(LED_STOPPED, LOW);
120     digitalWrite(LED_WALKING, HIGH);
121     digitalWrite(LED_RUNNING, LOW);
122 } else {
123     pace = "Running";
124     digitalWrite(LED_STOPPED, LOW);
125     digitalWrite(LED_WALKING, LOW);
126     digitalWrite(LED_RUNNING, HIGH);
127 }
128
129 // Check for reset button press
130 if (digitalRead(BTN_RESET) == LOW) {
131     displayMessage("Resetting...");
132     while (digitalRead(BTN_RESET) == LOW); // Wait for release
133     stepCount = 0;
134     displayMessage("Steps Reset");
135     delay(1000);
136 }
137
138
139 const char* getCurrentPace() {
140     // Determine pace based on current accelerometer reading
141     AxisData data = readAllAxes();
142     float mag = sqrt(data.x * data.x + data.y * data.y + data.z * data.z);
143     if (mag < 0.3) return "Stationary";
144     else if (mag < 2.0) return "Walking";
145     else return "Running";
146 }
147
```





Evaluation, feedback & reflection

Group Self-Evaluation

	Expert	Proficient	Developing	Novice
Overall Delivery and Presentation				
Team Profile, Collaboration & Communication				
System Overview				
Project Management				



Incorporation of improvement following feedback

Criteria	Feedback Received	Action Taken
Evaluation, feedback & reflection	Provide a little introduction to the project before the technical/detailed portion of the presentation	Added a concise project introduction at the beginning of the presentation to give context before diving into technical details.
Schematic Design	Concern about manually wiring LiPos in series	Replaced with professionally matched 2S LiPo with built-in balancing and protective circuitry.
	—	Accelerometer signal wiring and ADC channel assignment verified and documented in signal mapping table.
	Cutoff frequency not mentioned	Cutoff frequency now calculated and explicitly stated in both report and presentation.
	—	GPIO usage clarified in the signal mapping table, with connections visualised in schematic and layout.
	—	Final signal mapping table prepared using actual net names for traceability.
Embedded Software Design	—	Final version includes flowchart showing main loop and all subroutine interactions.
	—	Implemented self-test on startup, with status shown via onboard LEDs.
	—	Added calibration using ST pin input, with prompts on OLED.
	—	Step detection refined with signal filtering and adaptive thresholding.
	Needs to be better rehearsed	Clarified the logic, rehearsed explanation, and ensured confident delivery during presentation.



Q&A

Thank you for joining us today!